

Literature answer: endpoint real interpolation of Whitney-averaged tent spaces

Status: literature_already_answered (affirmative answer)

Original question

P. Auscher, S. Bechtel, and L. Haardt, *A new scale of function spaces characterizing homogeneous Besov spaces*, arXiv:2512.07399, Section 4, page 40, immediately before Lemma 4.3, identify an endpoint omitted from their tent-space interpolation theorem. Their scale $T_\beta^{p,q,r}$ was defined only for $0 < p < \infty$, and the question is whether $Z_\beta^{\infty,q,r}$ can likewise be obtained by real interpolation of endpoint tent spaces. The source snapshot used by the queue described this as future work.

Decisive later theorem

L. Haardt, *A coherent theory of tent spaces and homogeneous Triebel–Lizorkin spaces*, arXiv:2602.20305v2, first defines $T_\beta^{\infty,q,r}$ by the Carleson-box quasi-norm

$$\sup_{\tau > 0} \sup_{y \in \mathbb{R}^d} \left(\int_0^\tau \int_{B(y,\tau)} \left(\int_{t/2}^t \int_{B(x,t)} |s^{-\beta} f(s,z)|^r dz ds \right)^{q/r} \frac{dx dt}{t} \right)^{1/q},$$

with the usual modifications for infinite exponents.

Proposition 3.31, page 51, proves that whenever $0 < p \leq \infty$, $0 < q, q_0, q_1, r \leq \infty$, $\beta_0 \neq \beta_1$, and $0 < \theta < 1$,

$$(T_{\beta_0}^{p,q_0,r}, T_{\beta_1}^{p,q_1,r})_{\theta,q} = Z_\beta^{p,q,r}, \quad \beta = (1 - \theta)\beta_0 + \theta\beta_1,$$

with equivalent quasi-norms. Taking $p = \infty$ is exactly the missing endpoint characterization.

Identification and proof route

The supporting paper explicitly says immediately before Proposition 3.31 that the characterization of $Z_\beta^{\infty,q,r}$ was missing and that the proposition supplies it. Its proof extends the nesting argument used in the source paper: endpoint tent spaces sit between the corresponding Z-spaces with the scale exponents $\min\{p, q_i\}$ and $\max\{p, q_i\}$, and the full-range real interpolation theorem for Z-spaces identifies both outer interpolation spaces with the same $Z_\beta^{p,q,r}$.

The current PDF of arXiv:2512.07399 now replaces the future-work sentence by a direct pointer to Proposition 3.31 of the supporting paper. This is therefore an explicit later-paper answer, not a new result of the present run.

References

- [1] P. Auscher, S. Bechtel, and L. Haardt, *A new scale of function spaces characterizing homogeneous Besov spaces*, arXiv:2512.07399.
- [2] L. Haardt, *A coherent theory of tent spaces and homogeneous Triebel–Lizorkin spaces*, arXiv:2602.20305v2 (2026), Proposition 3.31.